Alpaca accounts: the scheduled fetch

The second way trades enter the journal. Add an Alpaca account on the page and the journal reads its fills on a schedule, pairs them into round trips and writes them in. This is the journal's best-built part: the pairing is the model a rebuild should use for every source.

Written from `journal/src/addons/accounts.js`, `journal/src/addons/fillsToTrades.js` and the accounts panel in `journal/public/addons.js`. Covered by `tests/journal.alpacaAccounts.test.js` (18 tests) and `tests/journal.import.test.js` (19 tests, the pairing).

The panel (Trades tab, above the trade list)

Add an Alpaca account: name, API key id, secret (a password field), paper or live, *every N min*, *last N days*, **add account**. After saving, the message says the first fetch comes within a minute.

Each account is one row:

Shown	Meaning
name	as typed
`paper` or `LIVE` (amber) · key id · every N min · last N days · `PAUSED`	the settings; the key id is shortened to its first 4 and last 2 characters
status line	`not fetched yet`, or `last fetch <time> ET · F fills → T trades (A new, U updated)`, or `last fetch <time> ET FAILED: <reason>` in red
fetch now	fetches at once, then reloads the trades
test key	asks Alpaca for the account: `key works · account <number> · <status>` or the reason it failed
pause / **resume**	turns the schedule off or on
remove	asks for a second tap (`remove? its trades stay`); the account goes, its trades stay

The panel re-reads the list every minute and whenever the page comes back into view, so a scheduled fetch shows without a reload. Every account's name is also added to the page's account filter, so an account that has not traded yet can still be chosen.

The settings and their checks

Field	Rule	Default
name	1 to 40 characters; unique, ignoring case	—
key id, secret	both or neither; no quotes or spaces; key ≥ 10 and secret ≥ 20 characters. On an edit, leaving both out keeps the old pair	—
paper	paper uses `https://paper-api.alpaca.markets`, live `https://api.alpaca.markets`	paper
every	5 to 1440 minutes, rounded	30
look back (`days`)	1 to 90 days, rounded	5
enabled	on or off	on

An account gets a random 12-hex-character id.

Where the keys live

`<JOURNAL_DATA_DIR>/journal-accounts.json`, normally `data/journal/`.

- Written to a temporary file, then renamed over the old one, mode **0600**.
- Gitignored by name. The nightly backup does **not** take it: the backup copies only `journal.db` from that folder.
- No answer from the server ever carries a secret. The page receives `id, name, paper, enabled, everyMinutes, days`, the shortened key id, `hasKeys` and `lastRun`.
- The only Alpaca calls made with these keys are `GET /v2/account` (test key) and `GET /v2/account/activities/FILL` (the fetch). The keys are used read-only.

The same file is also the **last fallback** for the market-data key the technical compute uses (ANALYSIS.md).

The schedule

- A timer ticks **every 60 seconds**.
- Each tick takes every account that is enabled, has keys, and whose last run (success **or** failure) is at least *every* minutes old, and fetches them **one at a time**.

- A failed fetch is recorded with its reason and waits for its next turn. It does not retry every minute.
- Only one fetch per account can run at once; a second one answers `a fetch for this account is already running`.

One fetch

1. `since` = now − *days*.
2. Ask Alpaca for every fill after `since`, oldest first, 100 per page, up to **50 pages (5,000 fills)**. Each fill keeps `id, symbol, side, qty, price, transaction_time`.
3. Pair the fills into trades (next section).
4. Write the trades (the section after).
5. Save `lastRun = { at, ok, fills, trades, added, updated, error }` on the account.

Fills into trades

`tradesFrom(fills, accountId, accountName)`: fills are grouped by symbol and sorted by time (pages do not guarantee order). Then, per symbol, a **running position** is tracked:

Position	Becomes
0 → not 0	a trade opens; its direction is this fill's (`buy…` is long, anything else short)
moves further the same way	more entry; the entry price is the **volume-weighted average** of every entry fill
moves back toward 0	exit; the exit price is the volume-weighted average of every exit fill
reaches 0	the trade is finished
crosses 0 in one fill	the fill is **split**: the part up to 0 closes the first trade, the rest opens a new trade the other way
not 0 at the end of the window	the trade is kept as **open**, with no exit

A trade counts as closed only when its exit shares reach its entry shares; a partly closed trade is sent as open with no exit price. Times are taken in **New York**: the trade's `date` is the entry's New York date, and entry and exit times are `HH:MM:SS` New York clock.

The id is `alpaca:<accountId>:<first fill's Alpaca id>`. The account is part of the id, so two accounts trading the same stock at the same second can never overwrite each other.

Writing them

For each trade:

Case	Action
a field is missing (id, date, ticker, direction, shares, entry price)	skipped
a trade with this id exists and is **closed**	skipped: **a closed trade is never rewritten**, because by then it may carry a setup or a note
a trade with this id exists and is open	updated: shares, entry, exit, P&L, % move, duration, status, account
new	inserted

Stored values: `commission = 0`, `net_pnl = gross_pnl`, `pct_move` signed by direction, `source = alpaca`, `account` = the account's **name**, `entry_time` `00:00:00` when missing, `duration_ms` from the two clock times on the entry date. Every trade that is closed after the write is offered to the grading copy (ANALYSIS.md).

Limits found while checking (recorded in KNOWN-ISSUES.md)

	What happens
A position older than the window (D-J11)	Fills before `since` are not fetched. A long bought 7 days ago and sold today, with *days* = 5, arrives as one sell: it opens a **short** that stays open. The real trade, if an earlier fetch stored it as open, keeps its old id and is never closed. **Tested.** Any trade held longer than *days* hits this
Swing trades (D-J12)	A trade has one date (the entry's) and two clock times. Bought Monday 10:00 and sold Wednesday 11:30 is stored as exit 11:30 with a duration of **1.5 hours**. **Tested**
No fees (D-J15)	Commission is always 0. Alpaca stock trades have no commission, but regulatory and pass-through fees are not read
Renaming an account	Trades carry the account's name. Open trades take the new name on the next fetch; closed trades keep the old one, so one account's history shows under two names
5,000 fills	A fetch reads at most 50 pages; an account with more fills in its window loses the newest

What a rebuild should keep

The pairing itself: running position, weighted prices, reversals split, open trades kept, a stable id that includes the account. It should be the only pairing code, used for files too. Add to it: fetch from **before** the oldest open position (or carry the position across fetches), give a trade an exit date, and read fees.